

VS Code interface showing a Python script for training a neural network. The Explorer pane on the left shows the project structure:

- heart.csv
- neural.py (2)
- nn_model.pkl
- predict.py
- thal_encoder.pkl
- x_scaler.pkl

The main editor displays the code in `neural.py`:

```
22 model = Sequential()
23     Dense(16, input_shape=(x_train.shape[1],), activation='relu'),
24     Dense(8, activation='relu'),
25     Dense(4, activation='relu'),
26     Dense(1, activation='sigmoid')
27 )
28
29 model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
30 model.fit(x_train, y_train, epochs=50)
31 loss, accuracy = model.evaluate(x_test, y_test, verbose=0)
32 print("Accuracy:", accuracy)
33 print("Loss:", loss)
34
35 y_prob = (model.predict(x_test))
36 y_pred = (y_prob >= 0.5).astype(int)
37 print("Classification Report:")
38 print(classification_report(y_test, y_pred))
39 print("Confusion Matrix:")
40 print(confusion_matrix(y_test, y_pred))
41
42 joblib.dump(model, 'nn_model.pkl')
43 joblib.dump(x_scaler, 'x_scaler.pkl')
44 joblib.dump(thal_encoder, 'thal_encoder.pkl')
45 print("Model, scaler, and encoder saved successfully.")
```

The TERMINAL pane shows the execution output:

```
PS D:\Coding> & "C:\Program Files\Python312\python.exe" d:/Coding/neural.py
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
Epoch 1/50
8/8 ----- 1s 4ms/step - accuracy: 0.4669 - loss: 0.7125
Epoch 2/50
8/8 ----- 0s 4ms/step - accuracy: 0.5579 - loss: 0.6925
Epoch 3/50
8/8 ----- 0s 3ms/step - accuracy: 0.6033 - loss: 0.6748
Epoch 4/50
8/8 ----- 0s 3ms/step - accuracy: 0.6777 - loss: 0.6574
Epoch 5/50
8/8 ----- 0s 3ms/step - accuracy: 0.7066 - loss: 0.6401
Epoch 6/50
8/8 ----- 0s 3ms/step - accuracy: 0.7273 - loss: 0.6226
Epoch 7/50
8/8 ----- 0s 3ms/step - accuracy: 0.7273 - loss: 0.6040
Epoch 8/50
8/8 ----- 0s 3ms/step - accuracy: 0.7273 - loss: 0.5844
Epoch 9/50
```

The status bar at the bottom indicates: Ln 38, Col 45 - Spaces: 4 - UTF-8 - CRLF - Python - Python 3.12 (64-bit)

EXPLORER

CODING

heart.csv

neural.py 2

nn_model.pkl

predict.py

thal_encoder.pkl

x_scaler.pkl

neural.py

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PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS D:\Coding> & "C:\Program Files\Python312\python.exe" d:/Coding/neural.py
8/8 0s 3ms/step - accuracy: 0.8843 - loss: 0.2742
Accuracy: 0.8524590134620667
Loss: 0.33205196261405945
2/2 0s 28ms/step
Classification Report:
      precision    recall  f1-score   support

     0       0.91      0.89      0.90        44
     1       0.72      0.76      0.74        17

 accuracy          0.85          61
  macro avg       0.81      0.83      0.82          61
 weighted avg     0.86      0.85      0.85          61

Confusion Matrix:
[[ 39   5]
 [  4 13]]
Model, scaler, and encoder saved successfully.
PS D:\Coding>
```

OUTLINE

TIMELINE

Ln 38, Col 45 Spaces: 4 UTF-8 CRLF Python Python 3.12 (64-bit)

VS Code interface showing a Python script for heart disease prediction.

EXPLORER

- CODING
 - .vscode
 - heart.csv
 - neural.py
 - nn_model.pkl
 - predict.py
 - thal_encoder.pkl
 - x_scaler.pkl

predict.py

```
13 restecg = int(input("Enter Rest ECG (0-2): "))
14 thalach = int(input("Enter Max HR: "))
15 exang = int(input("Enter Exer Angina (1/0): "))
16 oldpeak = float(input("Enter ST Depression (oldpeak): "))
17 slope = int(input("Enter Slope (1-3): "))
18 ca = int(input("Enter Vessels(0-3): "))
19 thal = input("Enter Thal (normal/fixed/reversible): ")
20
21 input_data = pd.DataFrame([[age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal_encoder.transform([thal])[0]]],
22 | columns=['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal_encoded'])
23
24 input_data_scaled = x_scaler.transform(input_data)
25 prediction_prob = model.predict(input_data_scaled)[0][0]
26 if prediction_prob >= 0.5:
27 | prediction = 1
28 else:
29 | prediction = 0
30
31 print("Prediction Probability:", prediction_prob)
32 if prediction == 1:
33 | print("Result: Heart Disease Detected")
34 else:
35 | print("Result: No Heart Disease")
```

TERMINAL

```
PS D:\Coding> & "C:\Program Files\Python312\python.exe" d:/Coding/predict.py
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not
be used. Please use WSL2 or the TensorFlow-DirectML plugin.
Enter Age: 20
Enter Sex (1-Male, 0-Female): 1
Enter Chest Pain (0-4): 1
Enter Resting BP(mmHg): 146
Enter Cholesterol: 247
Enter FBS (1/0): 1
Enter Rest ECG (0-2): 1
Enter Max HR: 150
Enter Exer Angina (1/0): 0
Enter ST Depression (oldpeak): 2
Enter Slope (1-3): 2
Enter Vessels(0-3): 1
Enter Thal (normal/fixed/reversible): normal
1/1 0s 64ms/step
Prediction Probability: 0.2922414
Result: No Heart Disease
PS D:\Coding>
```

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